

Forensics for Cybercrimes Committed Using Artificial Intelligence

A Digital Forensic Framework for AI-Assisted Cybercrime

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2026

Research Context

Artificial Intelligence is changing the cybercrime landscape.

Traditional Cybercrime

- ▶ Human-controlled systems
- ▶ Identifiable devices
- ▶ Conventional malware
- ▶ Established forensic procedures

AI-Assisted Cybercrime

- ▶ Automated attacks
- ▶ Synthetic identities
- ▶ AI-generated content
- ▶ Autonomous decision-making

The Forensic Challenge



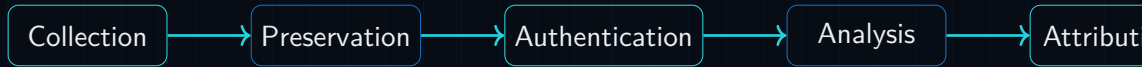
The central question is not merely:

What happened?

but also:

Who or what caused it, and can we prove it?

Forensic Evidence Pipeline



Formalising the Evidence Problem

Let

$$E = \{e_1, e_2, \dots, e_n\}$$

represent the set of collected digital evidence.

Let

$$A \subseteq E$$

represent evidence that has successfully passed an authentication process.

Therefore,

$$A \subseteq E$$

and

$e_i \in A \Rightarrow e_i$ satisfies the defined authentication criteria.

Research Objectives

1. Examine how AI can facilitate cybercrime.
2. Identify limitations in existing digital forensic approaches.
3. Investigate evidence provenance and integrity.
4. Develop a forensic framework for AI-assisted cybercrime.
5. Evaluate the proposed framework.

Expected Contribution



Conclusion

Central Argument

AI changes not only how cybercrimes can be committed, but also how digital evidence must be interpreted, authenticated and attributed.

AI + Cybercrime + Digital Forensics + Evidence

Thank You

Questions & Discussion

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